

Certificate I: The Hurwitz Rung $N = 7$

The cyclic cubic of $2 \cos(2\pi/7)$, the Cardano pairing, and the $\pi/7$ phase

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The “Dynamic Principle” program — the hodge-laboratory repository

Standalone theorem monograph · Monograph 17/20

1 Setting

With the program extended to the four rungs $N = 7, 9, 15, 30$ (roadmap v1.3), the ladder acquired two cubic foundations: the Hurwitz rung $N = 7$ and the Macbeath rung $N = 9$. Certificate I is the first of the two new anchors: it secures, at level $N = 7$, the entire pipeline of the program — the conductor census of eigencharacters, the exact cubic layer of $b_{\text{Ch}}(7)$ in $\mathbb{Z}[\zeta]/(\Phi_7)$, and the Cardano closed form in the casus irreducibilis regime.

The rung is named after Adolf Hurwitz: the polynomial $x^3 + x^2 - 2x - 1$ is the minimal polynomial of $2 \cos(2\pi/7)$, a generator of the maximal totally real subfield of the cyclotomic field $\mathbb{Q}(\zeta_7)$ — the very field that lives in the discriminant $\Delta = -343 = -7^3$ of the Klein quartic (monograph 16/20). Note, however, that the $N = 7$ stand certifies not the Klein quartic itself (which has its own “Klein” stand and monograph 16/20) but the Fermat-level pipeline at $N = 7$: the Fermat curve X_7 has genus 15, and every mechanism of the program — census, periods, Γ -monomials — operates at this level unchanged.

Theorem 1 (I1 — the census of the prime level). *For the Fermat curve X_7 every eigencharacter (a, b) with $1 \leq a, b, a + b \leq 6$ has conductor $d = 7$; the conductor census consists of a single class, $\{7: 15\}$, and $\sum_d h_d = 15 = (7 - 1)(7 - 2)/2 = g$. Both census schemes (the direct one and the Möbius one) produce the same histogram.*

Proof. The level $N = 7$ is prime, so for any eigencharacter (a, b) with $1 \leq a, b, a + b \leq 6$ we have $a, b \not\equiv 0 \pmod{7}$, hence $\gcd(7, a, b) = 1$ and the conductor is $d = 7 / \gcd(7, a, b) = 7$ — the only one. The number of such pairs (a, b) is the count of lattice points in the triangle $1 \leq a, b, a + b \leq 6$: for each a from 1 to 5 there are exactly $6 - a$ values of b , giving $\sum_{a=1}^5 (6 - a) = 5 + 4 + 3 + 2 + 1 = 15$. The Möbius scheme: $h_7 = c(7) - c(1) = 15 - 0 = 15$, where $c(k) = (k - 1)(k - 2)/2$ is the number of eigencharacters of level k . Both schemes are integer identities; their agreement is deterministic. $\square$

Theorem 2 (I2 — the cubic layer of $b_{\text{Ch}}(7)$). *The numbers $x_1 = \zeta + \zeta^{-1}$, $x_2 = \zeta^2 + \zeta^{-2}$, $x_3 = \zeta^3 + \zeta^{-3}$ with $\zeta = e^{2\pi i/7}$ have exact elementary symmetric polynomials*

$$s_1 = x_1 + x_2 + x_3 = -1, \quad s_2 = x_1x_2 + x_1x_3 + x_2x_3 = -2, \quad s_3 = x_1x_2x_3 = 1,$$

computed by pure integer arithmetic in $\mathbb{Z}[\zeta]/(\Phi_7)$; therefore x_1 is a root of the cubic $x^3 + x^2 - 2x - 1$, and this is its exact minimal polynomial over $\mathbb{Q}$.

Proof. Sum. $s_1 = \zeta + \zeta^6 + \zeta^2 + \zeta^5 + \zeta^3 + \zeta^4$ is the sum of all nontrivial powers of ζ , i.e. $s_1 = -1$ (the Φ_7 -sum, since $1 + \zeta + \dots + \zeta^6 = 0$).

Pairwise products. Expand in $\mathbb{Z}[\zeta]/(\Phi_7)$ using $\zeta^{-k} = \zeta^{7-k}$: $x_1x_2 = (\zeta + \zeta^6)(\zeta^2 + \zeta^5) = \zeta^3 + \zeta^6 + \zeta + \zeta^4$; $x_1x_3 = (\zeta + \zeta^6)(\zeta^3 + \zeta^4) = \zeta^4 + \zeta^5 + \zeta^2 + \zeta^3$; $x_2x_3 = (\zeta^2 + \zeta^5)(\zeta^3 + \zeta^4) = \zeta^5 + \zeta^6 + \zeta + \zeta^2$. Adding, each of the six powers $\zeta^1, \dots, \zeta^6$ occurs exactly twice: $s_2 = 2(\zeta + \zeta^2 + \dots + \zeta^6) = 2 \cdot (-1) = -2$.

Product. $x_1x_2x_3 = (\zeta^3 + \zeta^6 + \zeta + \zeta^4)(\zeta^3 + \zeta^4) = \zeta^6 + \zeta^7 + \zeta^9 + \zeta^{10} + \zeta^4 + \zeta^5 + \zeta^7 + \zeta^8 = \zeta^6 + 1 + \zeta^2 + \zeta^3 + \zeta^4 + \zeta^5 + 1 + \zeta = 2 + (\zeta + \dots + \zeta^6) = 2 - 1 = 1$.

By Vieta’s formulas the cubic with roots x_1, x_2, x_3 is $x^3 - s_1x^2 + s_2x - s_3 = x^3 + x^2 - 2x - 1$. Minimality: the Galois group $\text{Gal}(\mathbb{Q}(\zeta_7)^+/\mathbb{Q}) \cong (\mathbb{Z}/7)^\times / \{\pm 1\} \cong C_3$ acts by multiplication of exponents

by 2: $1 \mapsto 2 \mapsto 4 \equiv -3 \mapsto -1 \equiv 1$ — a cycle on the three classes $\{1, 6\}, \{2, 5\}, \{3, 4\}$; the action is transitive, so $[\mathbb{Q}(x_1) : \mathbb{Q}] = 3$. Fingerprint: modulo 2 the cubic $x^3 + x^2 - 2x - 1 \equiv x^3 + x^2 + 1$ has no root in $\mathbb{F}_2$ ($0 \mapsto 1, 1 \mapsto 1$) — it is irreducible, independently confirming the degree 3. $\square$

Theorem 3 (I3 — the Cardano form and the pairing). *In terms of $t = \frac{7}{54} + \frac{7\sqrt{-3}}{18}$ and $\bar{t} = \frac{7}{54} - \frac{7\sqrt{-3}}{18}$ the largest root of the cubic is*

$$2 \cos \frac{2\pi}{7} = \sqrt[3]{t} + \sqrt[3]{\bar{t}} - \frac{1}{3}, \quad b_{\text{Ch}}(7) = 1 - \cos \frac{2\pi}{7} = \frac{7}{6} - \frac{\sqrt[3]{t} + \sqrt[3]{\bar{t}}}{2},$$

and the moduli of the Cardano arguments are paired by the exact identity $|t|^2 = (7/9)^3$, so that the product of the cube roots is $\sqrt[3]{t} \cdot \sqrt[3]{\bar{t}} = 7/9$ exactly.

Proof. The substitution $x = y - \frac{1}{3}$ removes the quadratic term: $p = -\frac{7}{3}, q = -\frac{7}{27}$, and the cubic becomes $y^3 - \frac{7}{3}y - \frac{7}{27} = 0$. The Cardano discriminant is

$$\Delta = \frac{q^2}{4} + \frac{p^3}{27} = \frac{49}{2916} - \frac{343}{729} = \frac{49 - 1372}{2916} = -\frac{1323}{2916} = -\frac{1323}{2916} = -\frac{49}{108} < 0$$

— the classical *casus irreducibilis*: all three roots are real, yet expressible only through complex cube roots. Hence $\sqrt{\Delta} = \frac{7i\sqrt{3}}{18}$ and

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{\Delta}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\Delta}} = \sqrt[3]{\frac{7}{54} + \frac{7\sqrt{-3}}{18}} + \sqrt[3]{\frac{7}{54} - \frac{7\sqrt{-3}}{18}},$$

so $x_1 = y - \frac{1}{3}$ and $b_{\text{Ch}}(7) = 1 - \frac{x_1}{2} = \frac{7}{6} - \frac{\sqrt[3]{t} + \sqrt[3]{\bar{t}}}{2}$.

The pairing. $|t|^2 = \frac{49}{2916} + \frac{147}{324} = \frac{49+1323}{2916} = \frac{1372}{2916} = \frac{343}{729} = \left(\frac{7}{9}\right)^3$; therefore $\sqrt[3]{t} \cdot \sqrt[3]{\bar{t}} = \sqrt[3]{|t|^2} = \sqrt[3]{(7/9)^3} = 7/9$ — an exact integer identity (the “the Cardano pairing” comment in the laboratory code).

Viète form. $|t|^{1/3} = \sqrt{7}/3$ and $\arg t = \arctan(3\sqrt{3})$, so the principal cube root is $u = \sqrt[3]{t} = \frac{\sqrt{7}}{3}e^{i\theta}$ with $\theta = \frac{1}{3}\arctan(3\sqrt{3})$, and $2 \cos \frac{2\pi}{7} = 2 \operatorname{Re} u - \frac{1}{3} = \frac{2\sqrt{7}}{3} \cos\left(\frac{1}{3}\arctan(3\sqrt{3})\right) - \frac{1}{3}$ — the real trigonometric form (the trisection of the angle $\arctan(3\sqrt{3}) \approx 79.1^\circ$).

Isolation. The roots $x_1 \approx 2.24698, x_2 \approx -0.55496, x_3 \approx -1.69202$ are strictly ordered; the numeric check at working precision (dps 60) gives a relative deviation of the Cardano form from $1 - \cos(2\pi/7)$ equal to $2.1 \cdot 10^{-61}$ — agreement within working precision, pinning the largest root. $\square$

Remark 4. The phase $\pi/7$ anchors the quantum ladder of the program: item G6 of certificate G secures the phases $\pi/7, \pi/15, \pi/30$ as levels of the single construction $\delta = \pi/N$, and the isotypy $(1, 1, 0, 1)$ of the Klein quartic ties ζ_7 to its periods. The rung $N = 7$ is thus the arithmetic foundation of the whole heptadic branch.

Protocol data: Certificate I — the stand numbers

Genus $g = 15$; census $\{7: 15\}$ (both schemes); minimal cubic $x^3 + x^2 - 2x - 1$; Vieta $(s_1, s_2, s_3) = (-1, -2, 1)$ exactly in $\mathbb{Z}[\zeta]/(\Phi_7)$; the cubic is irreducible over $\text{GF}(2)$; Cardano form $b_{\text{Ch}}(7) = \frac{7}{6} - \frac{\sqrt[3]{t} + \sqrt[3]{\bar{t}}}{2}$, $t = \frac{7}{54} \pm \frac{7\sqrt{-3}}{18}$; the pairing $\sqrt[3]{t}\sqrt[3]{\bar{t}} = 7/9$; $b_{\text{Ch}}(7) = 0.376510198141266469474995115995760 \dots$; protocol relative errors: Cardano $2.0 \cdot 10^{-36}$, periods (spot sample) $1.0 \cdot 10^{-36}$ (dps 35); phase $\pi/7$.

Summary of results: summary

Certificate I accepted (5/5): the census of the prime level by two schemes, the exact integer Vieta data in $\mathbb{Z}[\zeta]/(\Phi_7)$, irreducibility over $\text{GF}(2)$, the Cardano form in the casus irreducibilis with the exact pairing $7/9$, and the phase $\pi/7$. The cubic foundation of the ladder is established.

Verification: how to reproduce

`python3 laboratory.py` → “Certificates → I”; “Stands → N=7”; batch mode: a scenario with the bch item ($N = 7$); `--check-baseline` cross-checks $g = 15$, the symmetric data $(-1, -2, 1)$ and the errors against the reference.